

Mathematical Foundations for Computing

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Module 1

Module 1: Sets and Set Theory

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Module 2

Module 2: Number Systems and Bases

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Module 3

Module 3: Functions and Their Properties

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Module 4

Module 4: Relations

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Lesson 4.1 Understanding the concept of relations

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Definition of a relation: relation versus function
Video • 8 min
- ✔

Definition of a relation: relation versus function
Practice Assignment • Grade: 100%
- ✔

Matrix and graph representations of a relation
Video • 10 min
- ✔

Matrix and graph representations of a relation
Practice Assignment • Grade: 100%
- ✔

The properties of a relation: reflexive, symmetric and anti-symmetric
Video • 14 min
- ✔

The properties of a relation: reflexive, symmetric and anti-symmetric
Practice Assignment • Grade: 100%
- ✔

Relation properties: transitivity
Video • 5 min
- ✔

Relation properties: transitivity
Practice Assignment • Grade: 100%

Lesson 4.2 Equivalence, and partial and total order relations

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Equivalence relations and equivalence classes
Video • 6 min
- ✔

Equivalence relations and equivalence classes
Practice Assignment • Grade: 100%
- ✔

Partial and total order
Video • 6 min
- ✔

Partial and total order
Practice Assignment • Grade: 100%
- ✔

Building and analysing different types of mathematical relations
Dialogue • 15 min

Lesson 4.3 Summary and Assessments

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Topic summary
Reading • 15 min
- Check your understanding: End of module 4
Graded Assignment • 20 min

Course Wrap-Up

- Mathematical Foundations for Computing:
Course Summary
Reading • 10 min
- The Archive Room: A Mathematical Escape
Mystery
Role Play • 15 min
- Course summary
Video • 1 min

Topic summary

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A relation R from a set A to a set B is a subset of the Cartesian product $A \times B$. That is, $R \subseteq A \times B$. If $(a, b) \in R$, we say that a is related to b by R .

Properties of relations:

- Reflexive: Every element is related to itself. $\forall a \in A, (a, a) \in R$
- Symmetric: If an element is related to another, then the second element is related to the first. $\forall a, b \in A, (a, b) \in R \implies (b, a) \in R$.
- Anti-symmetric: If an element is related to another and vice versa, then they must be the same element. $\forall a, b \in A, (a, b) \in R \text{ and } (b, a) \in R \implies a = b$.
- Transitive: If an element is related to a second element, which is related to a third element, then the first element is related to the third.

$$\forall a, b, c \in A, (a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

Adjacency matrix of a relation:

A relation can be defined by its adjacency matrix. Reflexivity, symmetry and anti-symmetry of a relation R can be easily deduced from its corresponding adjacency matrix.

Equivalence relation:

- A relation that is reflexive, symmetric and transitive.
- Partitions the set into equivalence classes.

Partial order:

- A relation that is reflexive, anti-symmetric and transitive.
- Provides a way to order elements in a set.

Total order:

- A partial order where every pair of elements is comparable.
- Also called a linear order.

Equivalence classes

Given an equivalence relation R on a set A , the equivalence class of an element $a \in A$, denoted by $[a]$, is:

$$[a] = \{x \in A \mid (a, x) \in R\}$$

Equivalence classes partition the set A into disjoint subsets where every element in A belongs to exactly one equivalence class.

By understanding relations and their properties, one gains a powerful tool for modelling and analysing complex systems in mathematics and computer science.

As you progress through your course, it is crucial to regularly assess your understanding and capabilities against the learning outcomes. This exercise is designed to help you reflect on your learning journey, identify areas where you may need to deepen your knowledge or skills, and develop a plan for improvement. Please follow the steps below:

- Review the learning outcomes.** For each of this topic's learning outcomes, rate your understanding on a scale of 1-5 (1 = very low, 5 = very high). Provide a brief explanation for your rating.
- Identify areas for improvement.** Review your ratings and explanations:
 - Identify any areas where you rated your understanding below a 4.
 - For each area identified, describe specific aspects you find challenging or unclear.
- Create an action plan.** For each identified knowledge gap, outline specific actions you will take to improve your understanding. Consider the following strategies:
 - Reviewing the course materials and textbooks.
 - Seek out additional resources that could aid your understanding.
 - Practicing additional problems or exercises.
 - Seeking help from the instructor or peers.
 - Scheduling additional study sessions on the topic.

Implement your action plan and regularly review your progress. Adjust your strategies as needed based on your ongoing self-assessment and any new feedback you receive. Remember, learning is a continuous process, and it's okay to revise your plan to better suit your needs.

Go to next item

✔ Completed